# Theater Ticketing System

# Software Requirements Specification

# V3

# 02/19/2026

Group 7
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Prepared for
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## Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

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Theater Ticketing System

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Theater Ticketing System

# 1. Introduction

The Theater Ticketing System is a web-based movie ticketing platform that is designed to support both online customers and in-person theater employees. In addition, it will also maintain synchronized, real-time ticket availability across all theaters. This Software Requirements Specification defines the functional and non-functional requirements necessary to design, implement, test, and deploy the system successfully.

The system will serve 20 theaters located in the San Diego area and operate within the Pacific Time Zone. It will replace the current slow and error-prone ticketing system with a more reliable, scalable, and secure solution.

## 1.1 Purpose

*The purpose of this SRS is to describe the functional and non-functional requirements of the Theater Ticketing System.*

This document provides a detailed description of the system's capabilities, interfaces, constraints, and expectations. It is supposed to be a reference for developers, project managers, stakeholders, and testers involved in the design, implementation, and maintenance of the system.

## 1.2 Scope

The Theater Ticketing System is a web-based application designed to allow customers to look at movies, select showtimes, reserve seats, and purchase tickets online.

The system will:

- Provide real-time seat availability
- Enable secure online payments
- Generate barcodes and QR codes for each respective ticket
- Send a confirmation email
- Provide tools for the owners so that they're able to manage each movie, its pricing, and schedules.
- Support ticket purchases up to two weeks in advance
- Support assigned seating for deluxe theaters and general admission for regular theaters
- Allow discount pricing for students, military personnel, and seniors

The system will not:

- Process refunds online
- Sell concessions
- Provide streaming services

The goal of the system is to streamline ticket purchasing, reduce wait times for purchasing tickets in person, and to also improve ease of changing movie parameters such as schedules and pricing.

## 1.3 Definitions, Acronyms, and Abbreviations

*This subsection should provide the definitions of all terms, acronyms, and abbreviations required to properly interpret the SRS. This information may be provided by reference to one or more appendixes in the SRS or by reference to other documents.*

This document follows standard IEEE SRS formatting conventions:

- "Shall" indicates a mandatory requirement.
- "Should" indicates a recommended requirement.
- "May" indicates an optional feature.
- Numbered sections are hierarchical and organized by feature category.
- Functional requirements are grouped under Section 3.2 and follow a structured format including inputs, processing, outputs, and error handling.

## 1.4 References

**PCI Security Standards Council.** (2024). *Payment Card Industry Data Security Standard: Requirements and Testing Procedures (Version 4.0.1)*. PCI Security Standards Council. https://docs-prv.pcisecuritystandards.org/PCI%20DSS/Standard/PCI-DSS-v4_0_1.pdf

## 1.5 Overview

The remaining sections of this SRS provide a general description, including characteristics of the users of this project, the product's hardware, and the functional and data requirements of the product. The general description of the project is discussed in section 2 of this document. Section 3 gives the functional requirements, data requirements and constraints and assumptions made while designing the Ticketing System. It also gives the user viewpoint of the product. Section 3 also gives the specific requirements of the product. Section 3 also discusses the external interface requirements and gives detailed description of functional requirements. Section 4 is for supporting information.

The remainder of this SRS is organized as follows:

- Section 2 provides a general description of the system, including user characteristics, constraints, and assumptions.
- Section 3 describes the specific functional and non-functional requirements of the system.
- Section 4 presents analysis models such as sequence diagrams, data flow diagrams, and state-transition diagrams.

- Section 5 describes the change management process for maintaining and updating system requirements.
- The appendices contain supporting documentation and additional reference material.

# 2. General Description

*This section of the SRS should describe the general factors that affect 'the product and its requirements. It should be made clear that this section does not state specific requirements; it only makes those requirements easier to understand.*

## 2.1 Product Perspective

The Theater Ticketing System is a web based system that uses other services such as payment systems as well as email services.

It will interact with: A database for data storage, a payment gateway so that people can pay with online services, an email system so that the ticketing system can send email confirmation, and other hardware such as ticket printers and barcode scanners.

## 2.2 Product Functions

The Major functions of the System will be:

- Account registration + login
- Movie Searching
- Time selection for each movie
- Seat selection with a chart of the theater
- Ticket reservation
- Cart for tickets
- Online payment processing
- Generation of tickets with ID and QR code
- Email confirmation
- Allow Owner to manage movies, pricing, and schedule

## 2.3 User Characteristics

Customers:

- Audience that will purchase tickets
- Basic knowledge of the internet
- Access to the ticketing system with either a web browser on a mobile or desktop device

Owner:

- Theater management or other staff
- Manage movies, schedules, and pricing
- Need secure login information

- Moderate knowledge of the internet

System Admin:

- Responsible for maintenance of system
- Manage database
- Advanced knowledge of the internet

## 2.4 General Constraints

- The system will use HTTPS for communications
- Payment processing will comply with PCI-DSS
- It will need a payment system such as Stripe
- The database must be relational
- It will comply with privacy regulations
- Passwords will be stored using secure methods
- The system shall be compatible with major web browsers (Chrome, Edge, Safari, Firefox).
- The system shall operate within standard internet bandwidth conditions.
- The system shall not require specialized hardware for customers.
- Administrative users shall require secure authentication credentials.

## 2.5 Assumptions and Dependencies

Assumptions:

- The theater will have reliable internet connection so that people can buy tickets online
- Users have access to a web browser and will be able to connect to the internet
- The payment service will be operational
- The email provider will be able to reliably email ticket information to customers
- Theater staff will have the ability to use the administration tools
- The seating layout will be predetermined and will be consistent
- The information that the customer provides about their payment information will be correct
- The system will be hosted in an environment that has adequate capacity
- It is assumed that users have access to reliable internet connectivity.
- It is assumed that third-party services (payment gateway and email provider) remain operational.
- It is assumed that the hosting environment supports scalable concurrent user traffic.

Dependencies:

- Payment system: a third party payment system to handle transactions securely
- Email provider: external email service to send ticket confirmation
- Database: needs a database in order to store movie information, schedules, seat availability, user data, and transaction history
- Barcode/Qr code library: needs a system to generate barcodes and QR codes

- Hardware: ticket printers and/or barcode scanners for digital tickets.

# 3. Specific Requirements

# 3.1 External Interface Requirements

## 3.1.1 User Interfaces

- The system shall provide a secure web-based payment interface accessible via modern browsers (Chrome, Edge, Safari, Firefox).
- The interface shall display:
  - Reservation ID
  - Total payment amount
  - Payment method options (credit/debit card, supported digital wallets)
- The system shall display a confirmation message after payment approval or a failure message if payment is declined.

## 3.1.2 Hardware Interfaces

- The system shall operate on standard user devices (desktop, tablet, mobile).
- No specialized hardware shall be required for payment processing.

## 3.1.3 Software Interfaces

- The system shall integrate with a third-party payment gateway (e.g., Stripe, PayPal).
- The system shall interface with the reservation database to verify active reservations before processing payment.
- The system shall integrate with an email service to send payment confirmations.

## 3.1.4 Communications Interfaces

- All communication between the user and system shall use HTTPS.
- Payment transactions shall use encrypted API communication with the payment gateway.

## 3.2 Functional Requirements

*This section describes specific features of the software project. If desired, some requirements may be specified in the use-case format and listed in the Use Cases Section.*

### 3.2.1 Process Ticket Payment

#### 3.2.1.1 Introduction

This function processes customer payments for reserved movie tickets.

#### 3.2.1.2 Inputs

- Reservation ID
- Total payment amount
- Card number or digital wallet credentials
- Expiration date
- CVV
- Billing address

#### 3.2.1.3 Processing

- Validate required input fields.
- Verify reservation ID is valid and active.
- Confirm payment amount matches reservation total.
- Submit transaction request to payment gateway.
- Receive transaction approval or denial.
- Store transaction record in database.

#### 3.2.1.4 Outputs

- Payment confirmation message (success or failure).
- Transaction ID (if successful).
- Electronic receipt sent via email.

#### 3.2.1.5 Error Handling

- If required fields are missing, display error message.
- If reservation ID is invalid, cancel transaction and notify user.
- If payment is declined, notify user and allow retry.
- If gateway communication fails, log error and prompt retry.

### 3.2.2 Payment Authorization Validation

#### 3.2.2.1 Introduction

This function validates user payment information and authorizes the transaction before final processing.

#### 3.2.2.2 Inputs

- Card number or digital wallet credentials
- Expiration date
- CVV
- Billing address
- Reservation ID
- Total amount

#### 3.2.2.3 Processing

- Validate input format (card length, expiration date format, required fields).
- Verify billing information is complete.
- Confirm reservation ID is valid and active.
- Send authorization request to payment gateway.
- Receive authorization approval or denial.

#### 3.2.2.4 Outputs

- Authorization approval or denial message.
- Authorization code (if approved).

#### 3.2.2.5 Error Handling

- Display error if required fields are missing.
- Display error if card format is invalid.
- Display message if authorization is declined.
- Log failed authorization attempts for security review.

#### 3.2.3.1 Seat Reservation

This function allows customers to temporarily reserve selected seats for a specific showtime.

#### 3.2.3.2 Inputs

- Movie ID

- Showtime ID
- Selected seat numbers

#### 3.2.3.3 Processing

- Verify selected seats are available.
- Lock seats for a maximum of 10 minutes.
- Generate Reservation ID.
- Update database to reflect temporary hold.

#### 3.2.3.4 Outputs

- Reservation ID
- Confirmation message

#### 3.2.3.5 Error Handling

- If seats are unavailable, display error message.
- If session expires, release seats automatically.

### 3.2.4 Ticket Purchase Restrictions

This function will restrict customers from abusing the ticket system and balance the system to make it fair.

#### 3.2.4.1 Processing

- Customers shall not purchase more than 20 tickets per transaction.
- Tickets shall only be available 14 days prior to showtime.
- Ticket purchases shall be disabled 10 minutes after showtime begins.

#### 3.2.4.2 Error Handling

- Display a restriction message if limits are exceeded.

### 3.2.5 Bot Prevention

This function will prevent spam from occurring in the system and abuse of bots to get tickets.

#### 3.2.5.1 Processing

- The system shall implement CAPTCHA verification at checkout.
- The system shall detect rapid repeated purchase attempts.
- Suspicious IP addresses shall be temporarily blocked.

### 3.2.6 Discount Processing

#### 3.2.6.1 Processing

- The system shall verify discount eligibility category.
- The system shall apply discount percentage automatically.
- The system shall log discount usage.

### 3.2.7 Administrator Management

#### 3.2.7.1 Processing

- Add, update, or remove movies.
- Modify ticket pricing.
- Configure showtimes.
- Update seating layout.

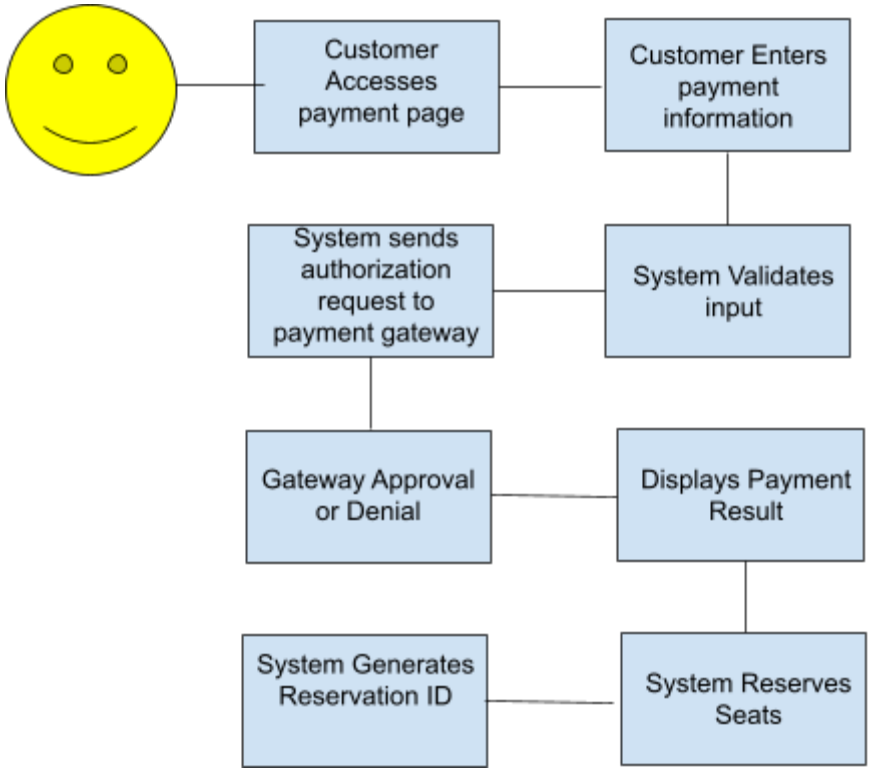
## 3.3 Use Cases

### 3.3.1 Complete Ticket Payment

**Actor:** Customer

1. Customer accesses payment page.
2. Customer enters payment information.
3. System validates input.
4. System sends authorization request to payment gateway.
5. Gateway returns approval or denial.
6. System displays payment result.
7. System stores transaction record and sends receipt (if approved).

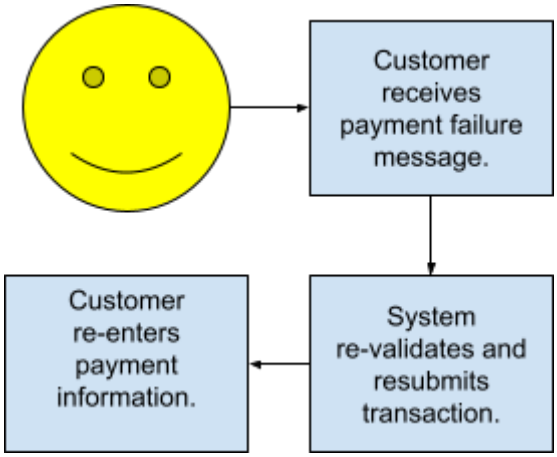
Theater Ticketing System




### 3.3.2 Retry Payment

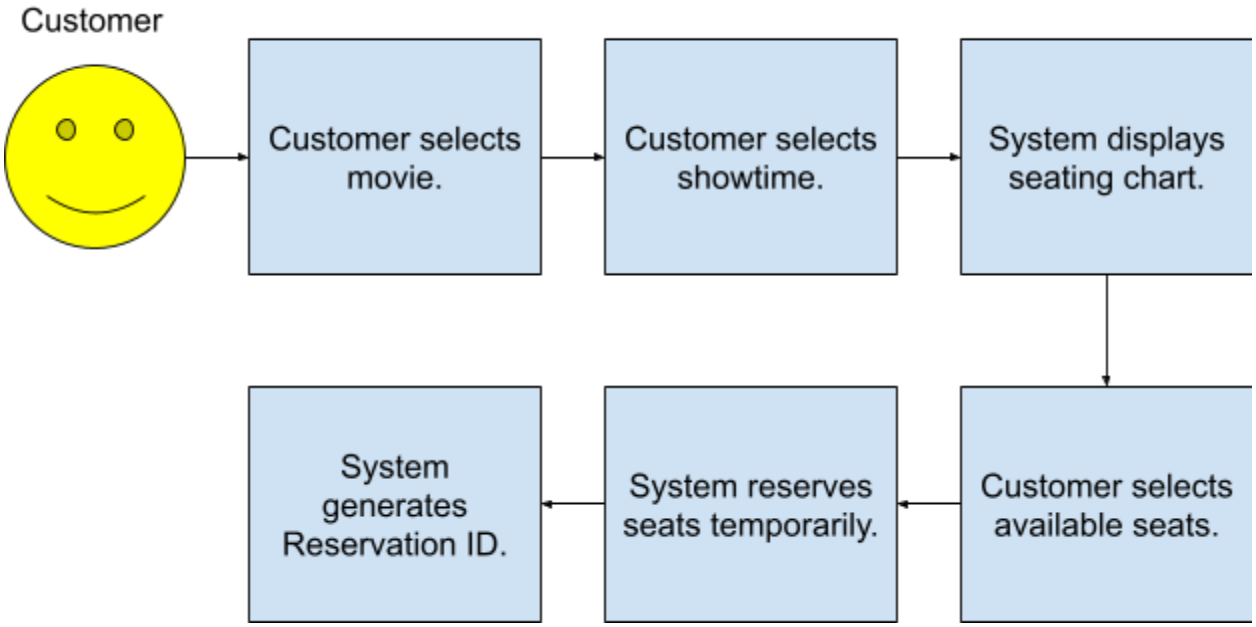
**Actor:** Customer

1. Customer receives payment failure message.
2. Customer re-enters payment information.
3. System re-validates and resubmits transaction.



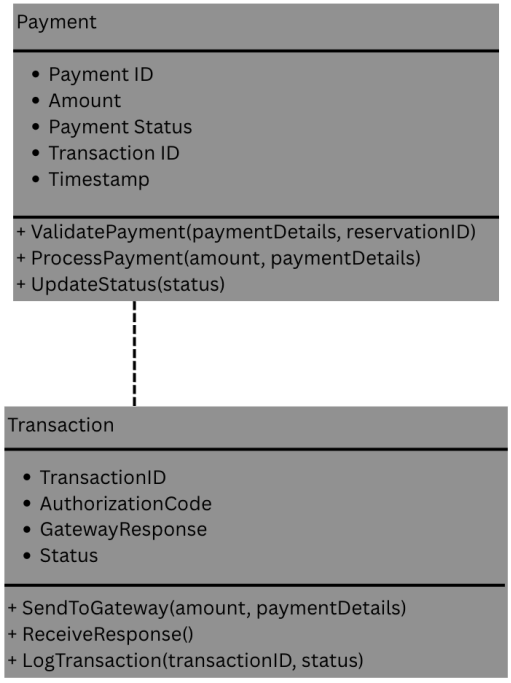
### 3.3.3 Reserve Seat

1. Customer selects movie.
2. Customer selects showtime.
3. System displays seating chart.
4. Customer selects available seats.
5. System reserves seats temporarily
6. System generates Reservation ID.



## 3.4 Classes / Objects

UML Diagram for Payment and Transaction Processes




### 3.4.1 Payment

#### *3.4.1.1 Attributes*

- PaymentID
- ReservationID
- Amount
- PaymentStatus
- TransactionID
- Timestamp

#### *3.4.1.2 Functions*

- ValidatePayment()
- ProcessPayment()

- UpdateStatus()

(References: 3.2.1, 3.2.2)

## 3.4.2 Transaction

### *3.4.2.1 Attributes*

- TransactionID
- AuthorizationCode
- GatewayResponse
- Status

### *3.4.2.2 Functions*

- SendToGateway()
- ReceiveResponse()
- LogTransaction()

(References: 3.2.1, 3.2.2)

## 3.4.3 Reservation

### *3.4.3.1 Attributes*

- ReservationID
- ShowtimeID
- SeatNumbers
- Status (Pending, Confirmed, Expired)
- Timestamp

### *3.4.3.2 Functions*

- LockSeats()
- ReleaseSeats()
- ConfirmReservation()

### *3.4.3.3 States*

- Pending
- Confirmed
- Expired

#### *3.4.3.4 Relationships*

- Associated with Payment
- Associated with Showtime

### 3.4.4 Movie

#### *3.4.4.1 Attributes*

- MovieID
- Title
- Description
- Duration
- Rating

#### *3.4.4.2 Functions*

- AddMovie()
- UpdateMovie()
- RemoveMovie()

### 3.4.5 Showtime

#### *3.4.5.1 Attributes*

- ShowtimeID
- MovieID
- TheaterID
- DateTime
- AvailableSeats

#### *3.4.5.2 Functions*

- UpdateAvailability()
- CloseShowtime()

## 3.5 Non-Functional Requirements

*Non-functional requirements may exist for the following attributes. Often these requirements must be achieved at a system-wide level rather than at a unit level. State the requirements in the following sections in measurable terms (e.g., 95% of transaction shall be processed in less than a second, system downtime may not exceed 1 minute per day, > 30 day MTBF value, etc).*

### 3.5.1 Performance

- 95% of payment transactions shall complete within 3 seconds.
- Authorization responses shall be processed within 2 seconds under normal operating conditions.
- The system shall support at least 1000 concurrent users without performance degradation.
- Seat reservation conflicts shall not exceed 2% under peak load.

### 3.5.2 Reliability

- The system shall maintain a 99% successful transaction processing rate, excluding user-declined payments.
- Failed transactions shall be logged for audit purposes.

### 3.5.3 Availability

- The payment processing module shall maintain 99% uptime per month.

### 3.5.4 Security

- All payment data shall be encrypted in transit using TLS 1.2 or higher.
- The system shall comply with PCI-DSS standards.
- The system shall not store full credit card numbers.
- The system shall not store CVV codes after transaction completion.
- The system shall implement CAPTCHA validation during checkout.
- Repeated failed transactions from the same IP address shall trigger temporary account or IP restriction.

### 3.5.5 Maintainability

- The payment gateway integration shall be modular to allow replacement without affecting other system components.
- System logs shall be maintained to assist troubleshooting and auditing.

### 3.5.6 Portability

- The payment processing module shall function on major web browsers.

- The system shall be accessible on Windows, macOS, Android, and iOS devices via browser.

## 3.6 Inverse Requirements

*State any *useful* inverse requirements.*

## 3.7 Design Constraints

*Specify design constrains imposed by other standards, company policies, hardware limitation, etc. that will impact this software project.*

- *Security constraints*
  - *Payment must be handled by a third-party gateway.*
- *Time constraints*
  - *Seat reservations will expire after 10 minutes if purchase isn't finished*
  - *Tickets become available 14 days before showtime*
- *Platform Constraints*
  - *Must run as a web-based system accessible by common web browsers*
- *Anti-bot-constraints*
  - *Must include CAPTCHA*

## 3.8 Logical Database Requirements

*Will a database be used? If so, what logical requirements exist for data formats, storage capabilities, data retention, data integrity, etc.*

*A relational database will be used to store users, movies, showtimes, seats, reservations, payments, transactions, discounts, and tickets*

- *Identifiers*
  - *Use UUID for primary values such as Movie, showtime, reservation, payment, transaction, ticket*
- *Money*
  - *Store currency values as amountCents: BIGINT*
- *Seat Representation*
  - *Seat numbers should be stored as strings such as A10 or combined as row:CHAR + seat: INT*
- *To prevent seats being double booked*
  - *Unique constraints should be used such as (showtime_id, seat_number for confirmed seats*
  - *Reservation expiration will automatically release seats*
- *Retention*
  - *Keep transaction records for refunds*
  - *Store only the payment gateway and not the actual card information*

## 3.9 Other Requirements

*Catchall section for any additional requirements.*

- *Email confirmation*
  - *After a successful purchase, system will send a confirmation email and receipt*
- *Usability*

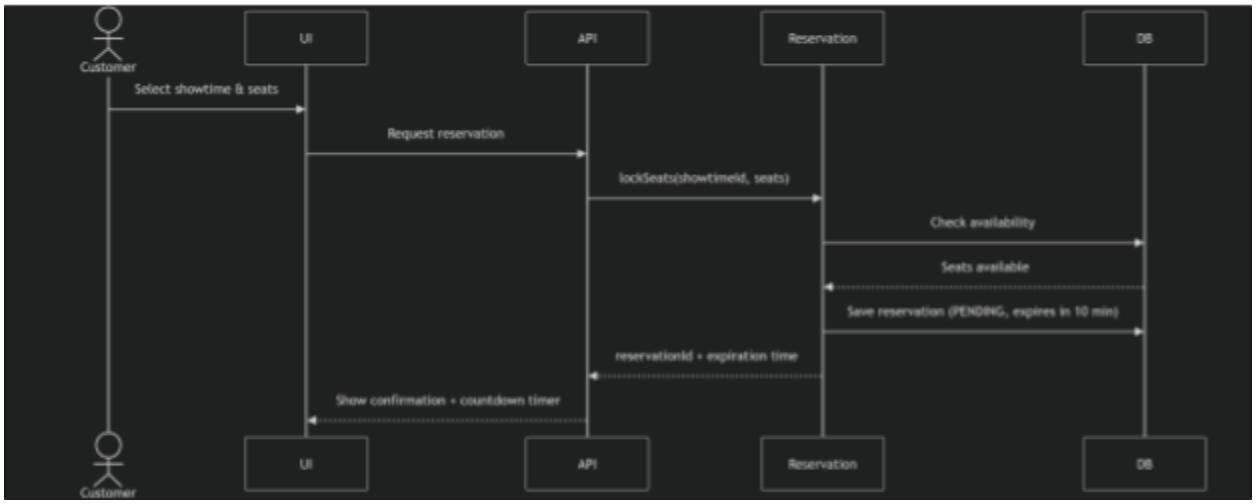
    - *Seat selection UI must show real-time availability as well as showing which seats are already taken clearly*
- *Logging/monitoring*
    - *Log checkout failures, suspicious activity, payment gateway erros and reservation expirations.*

# 4. Analysis Models

*List all analysis models used in developing specific requirements previously given in this SRS. Each model should include an introduction and a narrative description. Furthermore, each model should be traceable the SRS's requirements.*

## 4.1 Sequence Diagrams

- Seat Reservation




- Checkout and Payment

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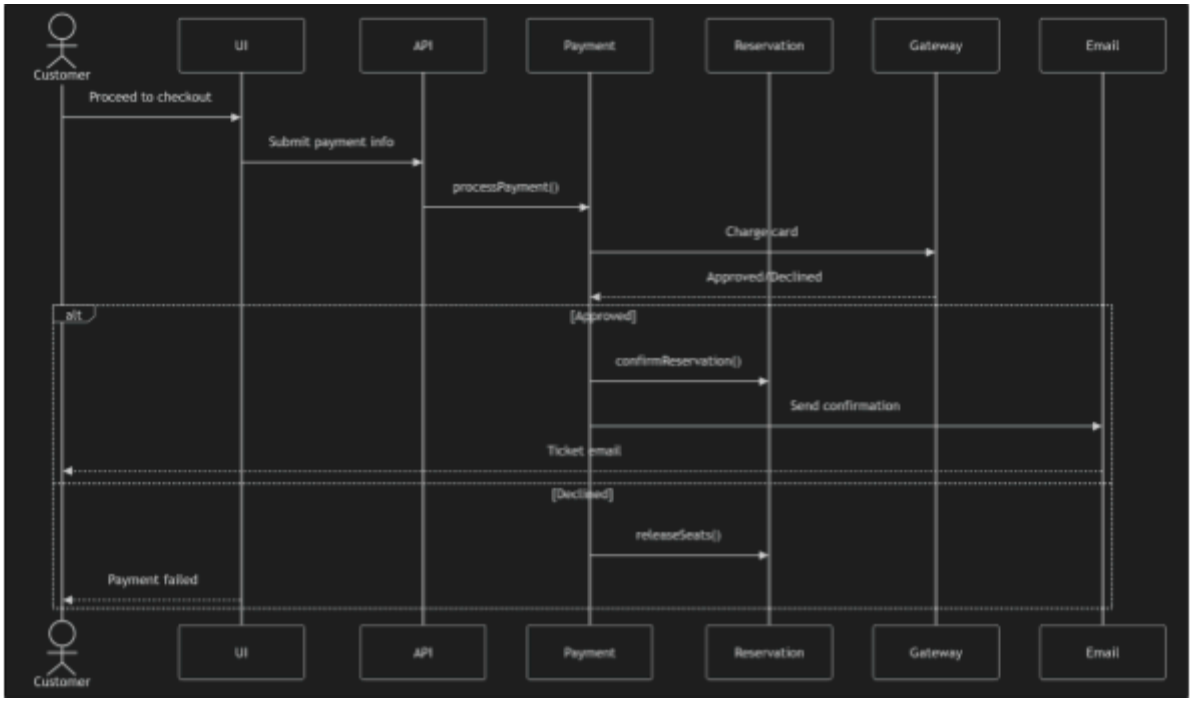
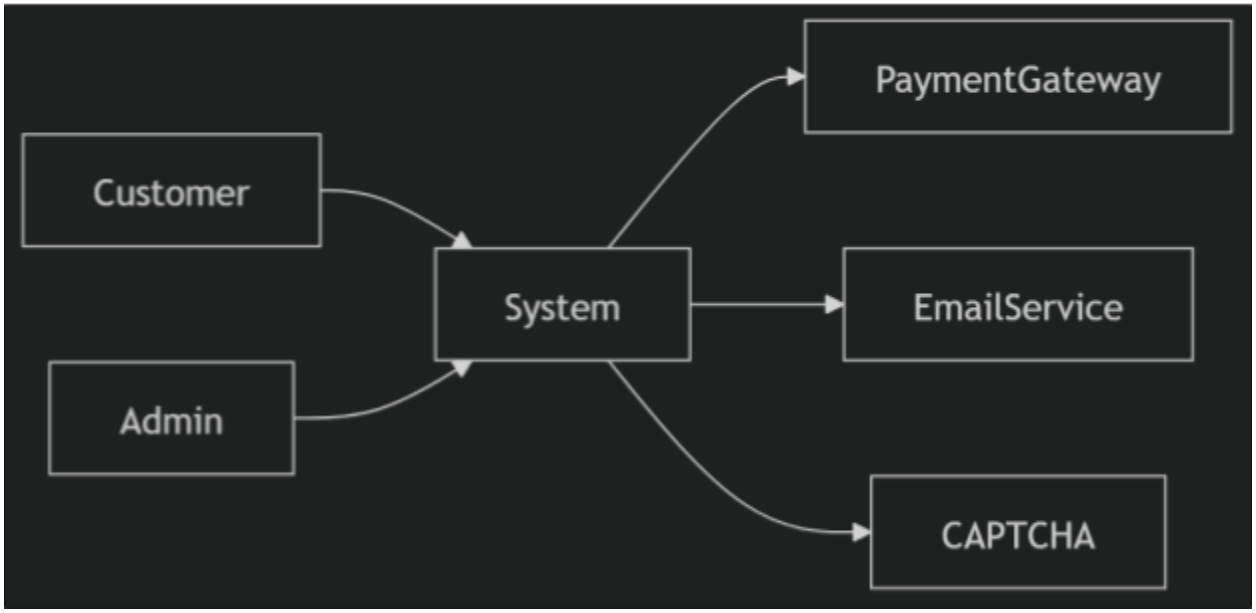
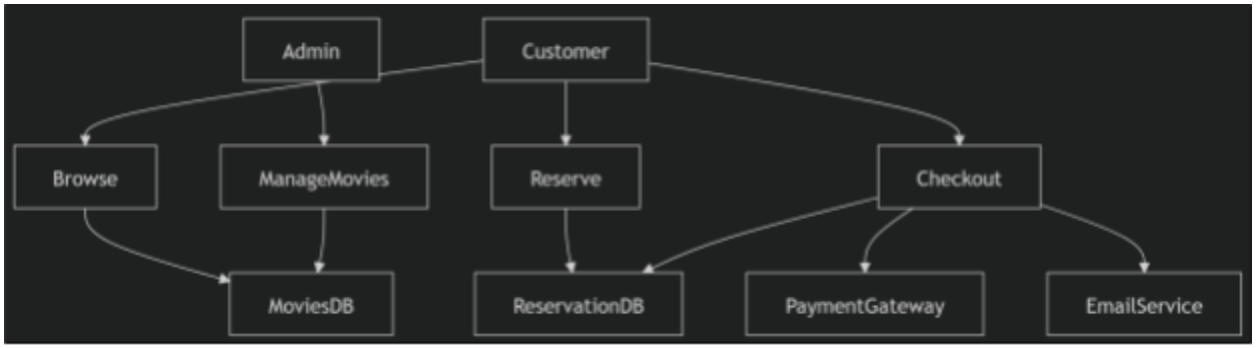



## 4.3 Data Flow Diagrams (DFD)

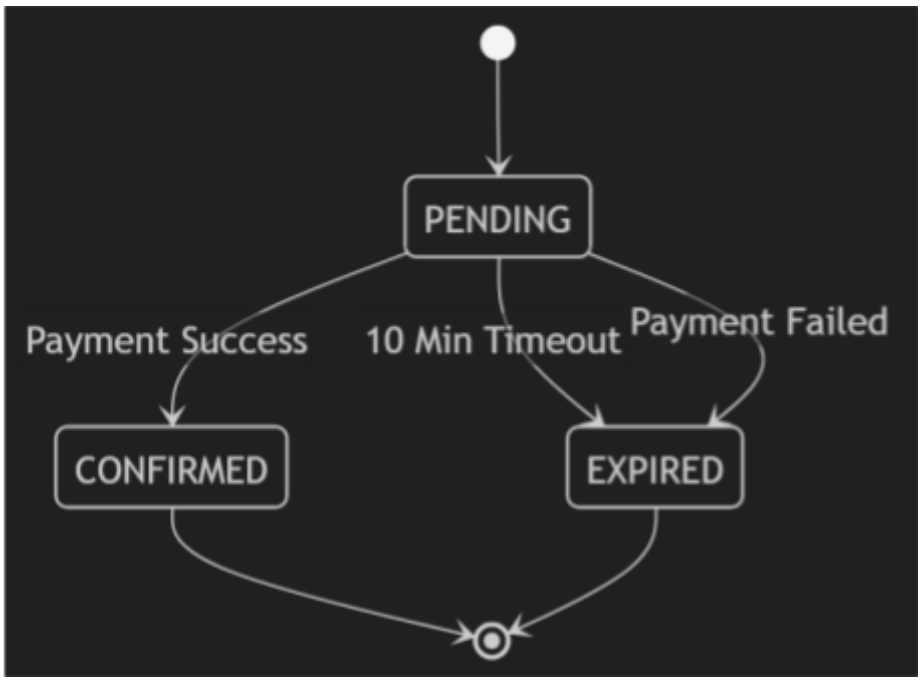


Theater Ticketing System

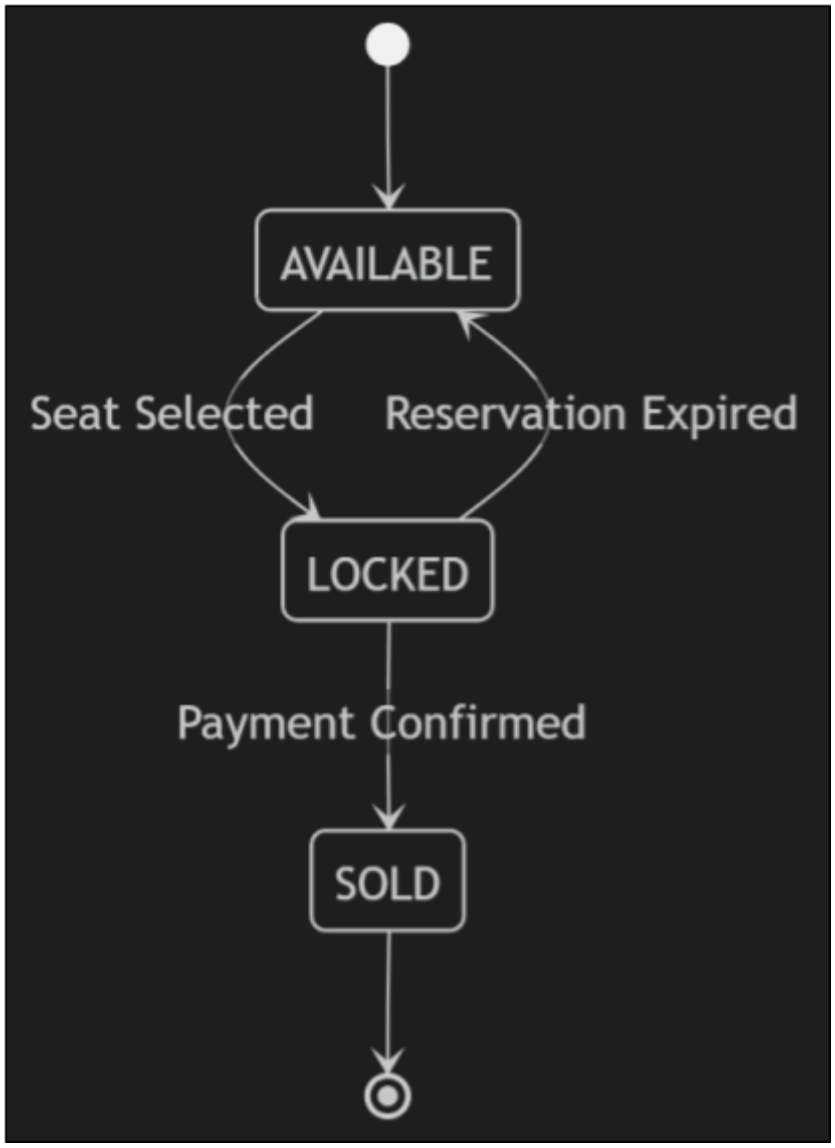



## 4.2 State-Transition Diagrams (STD)

Reservation Diagram




Seat Diagram




## 5. Change Management Process

*Identify and describe the process that will be used to update the SRS, as needed, when project scope or requirements change. Who can submit changes and by what means, and how will these changes be approved.*

## A. Appendices

*Appendices may be used to provide additional (and hopefully helpful) information. If present, the SRS should explicitly state whether the information contained within an appendix is to be considered as a part of the SRS's overall set of requirements.*

*Example Appendices could include (initial) conceptual documents for the software project, marketing materials, minutes of meetings with the customer(s), etc.*

## A.1 Appendix 1

## A.2 Appendix 2